

PULSE™ EXT 100

PC/ABS Engineering Resin

Overview PULSE EXT 100 engineering resin has been designed to provide excellent processability, high heat resistance, and high impact strength at both room and low temperatures.

PULSE EXT 100 engineering resin has been used in automotive exterior trim applications.

Physical	Nominal Value (English)	Nominal Value (SI)	Test Method
Density	1.14 g/cm ³	1.14 g/cm ³	ASTM D792
Melt Mass-Flow Rate (MFR) (230°C/3.8 kg)	1.3 g/10 min	1.3 g/10 min	ASTM D1238
Molding Shrinkage - Flow	6.0E-3 in/in	0.60 %	ASTM D955
Mechanical	Nominal Value (English)	Nominal Value (SI)	Test Method
Tensile Strength (Yield)	7500 psi	51.7 MPa	ASTM D638 ¹
Tensile Elongation (Break)	80 %	80 %	ASTM D638 ¹
Flexural Modulus	325000 psi	2240 MPa	ASTM D790
Flexural Strength (Yield)	12000 psi	82.7 MPa	ASTM D790
Impact	Nominal Value (English)	Nominal Value (SI)	Test Method
Notched Izod Impact			ASTM D256 ²
-20°F (-29°C), 0.126 in (3.20 mm)	9.0 ft-lb/in	480 J/m	
73°F (23°C), 0.126 in (3.20 mm)	12 ft-lb/in	640 J/m	
Instrumented Dart Impact			ASTM D3763 ³
-20°F (-29°C), 0.126 in (3.20 mm), Total Energy	530 in-lb	59.9 J	
73°F (23°C), 0.126 in (3.20 mm), Total Energy	480 in-lb	54.2 J	
Thermal	Nominal Value (English)	Nominal Value (SI)	Test Method
Deflection Temperature Under Load			ASTM D648
66 psi (0.45 MPa), Unannealed, 0.126 in (3.20 mm)	256 °F	124 °C	
264 psi (1.8 MPa), Unannealed, 0.126 in (3.20 mm)	225 °F	107 °C	
CLTE - Flow (-40 to 180°F (-40 to 82°C))	4.0E-5 in/in/°F	7.2E-5 cm/cm/°C	ASTM D696
Fill Analysis	Nominal Value (English)	Nominal Value (SI)	Test Method
Melt Density	1.02 g/cm ³	1.02 g/cm ³	
Melt Specific Heat	0.473 Btu/lb/°F	1980 J/kg/°C	ASTM C351
Melt Thermal Conductivity	1.8 Btu-in/hr/ft ² /°F	0.26 W/m/K	ASTM C177
No Flow Temperature	320 °F	160 °C	
Ejection Temperature	280 °F	138 °C	

Notes

These are typical properties only and are not to be construed as specifications. Users should confirm results by their own tests.

¹ Type I, 2.0 in/min (51 mm/min)

² 10 mil (0.25 mm) Notch Depth

³ 11.1 ft/sec (3.39 m/sec)

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North America		Europe/Middle East	
U.S. & Canada:	1-888-STYRON1		+800 444 11 444
	1-989-633-1718	Germany:	+31-11567-2626
Mexico:	+1-800-441-4369		+8001811361
Latin America		Asia Pacific	
Argentina:	+54-11-4319-0100		+800-7776-7776
Brazil:	+55-11-5188-9000		+603-7965-5319
Colombia:	+57-1-219-6000		
Mexico:	+52-55-5201-4700		

www.styron.com

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Published: August 2014

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